WORKED SOLUTION

STEP III 1989, Question 5

Pure • Chebyshev Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q5

The key idea

Put $x = \cosh\theta$. Then $e^{\pm\theta}$ are exactly $x \pm \sqrt{x^2-1}$, and $y = \cosh n\theta$ is the average of their nth powers. Expanding those powers with the binomial theorem, the odd powers of $\sqrt{x^2-1}$ cancel, so what is left is a polynomial in x. Once we have that polynomial in a tidy form, each coefficient is just a short sum of binomial coefficients.

Solution

The closed form

Let $\theta = \cosh^{-1} x$, so $\theta \geq 0$ and $x = \cosh\theta$. Then $\sinh\theta \geq 0$ and $\sinh^2\theta = x^2 - 1$, so $\sinh\theta = \sqrt{x^2-1}$. Hence

$$e^{\theta} = \cosh\theta + \sinh\theta = x + \sqrt{x^2-1}, \qquad e^{-\theta} = \cosh\theta - \sinh\theta = x - \sqrt{x^2-1}.$$

Therefore

$$y = \cosh n\theta = \frac{e^{n\theta} + e^{-n\theta}}{2} = \frac{(x+\sqrt{x^2-1})^n + (x-\sqrt{x^2-1})^n}{2},$$

as required.

Why y is an odd polynomial

Write $s = \sqrt{x^2-1}$. By the binomial theorem,

$$(x+s)^n + (x-s)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} s^j (1 + (-1)^j) = 2 \sum_{j \text{ even}} \binom{n}{j} x^{n-j} s^j.$$

The odd powers of s cancel, and every even power $s^{2m} = (x^2-1)^m$ is a polynomial in x. So

$$y = \sum_{m=0}^{\lfloor n/2 \rfloor} \binom{n}{2m} x^{n-2m} (x^2-1)^m.$$

Now take $n = 2k+1$, so m runs from 0 to k. Each term is an odd power $x^{2k+1-2m}$ times a polynomial in x^2, so it contains only odd powers of x, the highest being x^{2k+1}. Hence $y = a_0x + a_1x^3 + \cdots + a_kx^{2k+1}$.

To read off coefficients, put $j = k - m$ (so $j = 0, \ldots, k$) and use $\binom{2k+1}{2k-2j} = \binom{2k+1}{2j+1}$:

$$y = \sum_{j=0}^{k} \binom{2k+1}{2j+1} x^{2j+1}(x^2-1)^{k-j}.$$

The term with index j starts at x^{2j+1}. So x^{2r+1} comes only from the terms with $j \le r$, and from the jth of these we need the $(x^2)^{r-j}$ term of $(x^2-1)^{k-j}$, which is $\binom{k-j}{r-j}(-1)^{k-r}$. Hence

$$a_r = (-1)^{k-r} \sum_{j=0}^{r} \binom{2k+1}{2j+1}\binom{k-j}{r-j}. \qquad (*)$$

a_0

With $r = 0$, only $j = 0$ contributes:

$$a_0 = (-1)^k \binom{2k+1}{1} = (-1)^k(2k+1).$$

Part (i)

With $r = 1$, $j = 0, 1$ contribute:

$$\begin{aligned} a_1 &= (-1)^{k-1}\left[(2k+1)\binom{k}{1} + \binom{2k+1}{3}\right] = (-1)^{k-1}(2k+1)\left[k + \frac{2k(2k-1)}{6}\right] \\ &= (-1)^{k-1}(2k+1)\,k\,\frac{3+2k-1}{3} = (-1)^{k-1}\,\frac{2k(k+1)(2k+1)}{3}, \end{aligned}$$

as required.

Part (ii)

Here we need $k \ge 2$ for x^5 to appear. With $r = 2$, $j = 0, 1, 2$ contribute:

$$a_2 = (-1)^k\left[(2k+1)\binom{k}{2} + \binom{2k+1}{3}\binom{k-1}{1} + \binom{2k+1}{5}\right].$$

Each term has the factor $(2k+1)k(k-1)$; taking it out,

$$\begin{aligned} a_2 &= (-1)^k(2k+1)k(k-1)\left[\frac{1}{2} + \frac{2k-1}{3} + \frac{(2k-1)(2k-3)}{30}\right] \\ &= (-1)^k(2k+1)k(k-1)\,\frac{15 + (20k-10) + (4k^2-8k+3)}{30} \\ &= (-1)^k(2k+1)k(k-1)\,\frac{4(k+1)(k+2)}{30} = (-1)^k\,\frac{2(k-1)k(k+1)(k+2)(2k+1)}{15}. \end{aligned}$$

For the third term we used $\binom{2k+1}{5} = \frac{(2k+1)(2k)(2k-1)(2k-2)(2k-3)}{120} = \frac{(2k+1)k(k-1)(2k-1)(2k-3)}{30}$.

Ask yourself Our answer has a factor $(k+1)$ that the printed formula in (ii) lacks. Which one is right?

Test $k = 2$. Then $y = \cosh(5\theta)$ with $x = \cosh\theta$, and $\cosh 5\theta = 16x^5 - 20x^3 + 5x$, so $a_2 = 16$. Our formula gives $2 \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5/15 = 16$. The printed one gives $2 \cdot 1 \cdot 2 \cdot 4 \cdot 5/15 = \frac{16}{3}$, which can't be right because the a_r are integers. So the paper has a misprint: the factor $(k+1)$ is missing. The correct result is the one shown above.

The sum of the coefficients

Set $x = 1$ in the polynomial. Since the identity holds for $x \geq 1$ and $x = 1$ is allowed,

$$\sum_{r=0}^{k} a_r = y(1) = \cosh(n \cosh^{-1} 1) = \cosh 0 = 1.$$

Checks

- $k = 1$: $\cosh 3\theta = 4x^3 - 3x$. The formulas give $a_0 = -3$ and $a_1 = 2 \cdot 1 \cdot 2 \cdot 3/3 = 4$, and $-3 + 4 = 1$.
- $k = 3$: $\cosh 7\theta = 64x^7 - 112x^5 + 56x^3 - 7x$. The formulas give $a_0 = -7$, $a_1 = 56$ and $a_2 = -2 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 7/15 = -112$, and the coefficients sum to 1.
- We checked $(*)$ by computer against the recurrence $T_{m+1} = 2xT_m - T_{m-1}$ for every $k \leq 14$, and compared the polynomial with $\cosh(n \cosh^{-1} x)$ at random $x \geq 1$.

Common slips

- Forgetting that x^{2r+1} gets contributions from several terms of the expansion, not just one.
- Sign errors: every contribution to a_r carries the same sign $(-1)^{k-r}$, so factor it out first.
- Trying to force the algebra to reach the printed form of (ii). Testing a small case such as $k = 2$ shows it is misprinted.

Beyond STEP

The polynomial y is the Chebyshev polynomial $T_n(x)$, usually defined by $T_n(\cos\phi) = \cos n\phi$. Our hyperbolic version is the same identity for $x \geq 1$, and the two agree because a polynomial is fixed by its values on any interval. Chebyshev polynomials are central in approximation theory, since among monic polynomials of degree n, $2^{1-n}T_n$ has the smallest maximum size on $[-1, 1]$. They reappear in any first course in numerical analysis. For more STEP-style practice with this kind of algebra, try Siklos, *Advanced Problems in Mathematics*.

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